

RESEARCH

Open Access



Knowledge, attitudes, stigma perceptions, and practices towards HIV testing among youths aged 18–35 years in Kumba, Cameroon: a community-based cross-sectional study

Marie Clarie Fien Ndim^{1*†}, Frederick Nchang Cho^{1,2*†}, Diane Zinkeng Tongwa¹, Sih Ashley Kum¹, Christabel Afor Tatah¹, Franklin Ngwesse Ngome¹, Yvonne Mangwi Chi³, Jokwi Patrick Kofon^{2,4}, Edward Mbonigaba⁵, Eugene Mbuh Nyanjoh^{1†} and Andrew N. Tassang^{6†}

Abstract

Background Despite national efforts to scale up the human immunodeficiency virus (HIV) testing in Cameroon, many youths still remain untested. Understanding their knowledge, attitudes, stigma perceptions, and practices (KASP) towards HIV testing is vital for improving uptake. This study aimed to assess HIV-related KASP and identify barriers and facilitators to among youths aged 18–35 years in Kumba, Cameroon.

Methods A community-based cross-sectional study was conducted from February to April 2025 among 632 youths selected through a multistage sampling technique, stratified by municipalities and neighbourhoods across Kumba. Data were collected using a structured, pre-tested questionnaire. Descriptive statistics were used to summarise the sociodemographic data and KASP scores. Binary and ordinal logistic regression analyses were used to identify factors associated with HIV testing.

Results Out of the 632 participants, 528 (83.5%) demonstrated moderate knowledge of HIV testing, while 33 (3.6%) had high knowledge. Positive attitudes toward HIV testing were observed in 128 (20.3%) participants. Fear of receiving a positive result 222 (35.1%) and anticipated stigma 107 (16.9%) were cited as barriers to testing. Youths with no sexual partners were significantly more likely to have good knowledge (OR 2.98, 95% CI: 1.53–5.78, $p=0.001$), positive

[†]Marie Clarie Fien Ndim and Frederick Nchang Cho are joint first authors.

[†]Eugene Mbuh Nyanjoh and Andrew N. Tassang are joint senior authors.

*Correspondence:
Marie Clarie Fien Ndim
marieClariendim@gmail.com
Frederick Nchang Cho
nchang.cho@gmail.com

Full list of author information is available at the end of the article



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

attitudes (OR 3.15, 95% CI: 1.81–5.47, $p < 0.001$), and testing practices (OR 7.62, 95% CI: 4.05–14.32, $p < 0.001$). While stigma was associated with lower testing uptake, sociodemographic factors such as gender, education, and religion were not significantly associated with testing behaviour.

Conclusions Although knowledge and attitudes towards HIV were generally positive, substantial gaps remain in testing uptake. Fear of a positive result and stigma were key barriers cited by one-third and one-sixth of the participants, respectively. This highlights the need for targeted community-based interventions, including peer-led education programmes, stigma-reduction campaigns, and youth-friendly testing services. Integration of HIV education into youth outreach and health programmes is recommended. As a cross-sectional study, causal relationships could not be established.

Highlights - This study assessed HIV-related knowledge, attitudes, and testing practices among youths aged 18–35 years in Kumba, Cameroon. - Although general knowledge about HIV was high, misconceptions remained, and nearly one-third of the participants had never been tested for HIV. - HIV-related stigma, fear of positive results, and lack of youth-friendly services were major barriers to testing, underscoring the need for targeted awareness, improved counselling, and community-level interventions to promote regular voluntary HIV testing among youths in similar urban-rural communities.

Keywords Barriers to HIV testing, Community health, Health-seeking behaviour, HIV stigma, HIV testing, Knowledge, attitudes and practices, Youth

Background

Human immunodeficiency virus (HIV) and acquired immunodeficiency syndrome (AIDS) continue to be major public health challenges globally, with sub-Saharan Africa (SSA) bearing the highest burden. In 2023, approximately 39.9 million people were living with HIV worldwide, with 1.3 million new infections and 630,000 AIDS-related deaths reported [1–3]. The World Health Organisation (WHO) African Region accounts for about half of new global HIV infections, with an estimated adult prevalence of 3.4% among individuals aged 15–49 [3–5]. Youths aged 18–35 years contribute substantially to new infections, largely due to behavioural risk factors, socioeconomic vulnerabilities, and limited access to testing and prevention services.

In many parts of SSA, increased awareness of HIV has not translated into effective preventive practices or widespread uptake of testing services. In Cameroon, for example, HIV knowledge remains inconsistent, with only 28–41% of individuals aged 15–49 demonstrating a comprehensive understanding of HIV transmission, and only 55% of women aged 15–24 having ever tested for HIV [6]. This gap between awareness and action underscores persistent challenges in addressing HIV among youth.

Studies conducted across Cameroon reveal similar trends. In Fako Division, secondary school students showed moderate awareness of HIV, yet harboured misconceptions and displayed mixed attitudes towards people living with HIV/AIDS (PLWHA), with limited condom use and low testing rates [7]. In Kribi and Dschang, consistent condom use was reported by only 4.5–6.5% of participants, and a significant number believed in traditional cures for HIV [8, 9]. Within the Kumba Health District, stigma continues to affect

PLWHA, and comprehensive data on knowledge, attitudes, stigma perception, and practices among the general youth population aged 18–35 are lacking [10]. These findings show that cultural beliefs still shape how young people understand and respond to HIV.

Beyond knowledge, psychosocial factors such as fear and stigma significantly influence health-seeking behaviour. Fear of a positive diagnosis, perceived discrimination, and concerns about confidentiality remain major barriers for young people. In Uganda, internalised and anticipated stigma were identified as strong deterrents to HIV testing, particularly in rural populations [11]. Similarly, in Côte d'Ivoire and West Africa, studies report low levels of HIV prevention knowledge, with education and income emerging as predictors of testing uptake [12, 13].

Practical and structural barriers to testing also persist. Limited availability of youth-friendly services, poor staff attitudes, and social stigma continue to hinder access [14–16]. Additionally, sociodemographic factors such as age, media access, cultural norms, and healthcare infrastructure contribute to testing disparities. Musheke et al. [17] summarised these challenges in a systematic review, identifying personal fears, cost, and logistical constraints as key reasons for low HIV testing uptake in SSA.

Given these gaps, this study aimed to assess HIV-related knowledge, attitudes, stigma perceptions, and testing practices among youths aged 18–35 years in Kumba, Cameroon. Specifically, the study sought to: (i) assess the level of knowledge regarding HIV transmission, prevention, testing, and treatment; (ii) evaluate attitudes towards HIV testing, including perceptions of stigma, and willingness to test; (iii) examine actual testing practices, such as testing frequency, partner testing, and access to services; and (iv) identify sociodemographic

factors associated with HIV testing behaviours. Understanding these dimensions is critical for informing targeted, community-based interventions to improve HIV testing uptake and reduce the burden of new infections among youths in this high-prevalence setting.

Materials and methods

Study design and setting

A community-based cross-sectional study was conducted in Kumba I and Kumba II municipalities, Meme Division, Southwest Region of Cameroon, from February to April 2025. Kumba is the administrative capital of Meme Division, and is home to two health districts; Kumba North and Kumba South. Kumba I and II host the main urban and peri-urban populations, with a youth centre, religious institutions and informal settlements. These areas are characterised by high youth mobility, unemployment, and low use of health services, including HIV testing [18]. The prevalence of HIV in the Southwest Region stands at 2.8%, above the national average of 2.7% [19]. Several health facilities in Kumba provide HIV testing, but uptake remains suboptimal [19, 20]. Kumba is an important commercial and transit hub, with a population estimated at 144,413 as of 2024 [21]. This setting was selected for its high population of youths, known testing gaps, and potential for public health interventions targeting HIV prevention among young people.

Study population and eligibility criteria

Participants were male and female youths aged 18 to 35 years who had resided in Kumba I or Kumba II for at least six months prior to the study. Participants were eligible for inclusion if they were within the target age range, lived in either municipalities at the time of the study, and provided informed consent. Youths who were ill or unable to communicate at the time of data collection were excluded.

Sampling Method

A multistage sampling strategy was used to recruit participants:

First stage: Kumba I and Kumba II municipalities were randomly selected from the three municipalities in the city (Kumba I, II, and III).

Second stage: Due to the Anglophone crisis, 13 quarters were selected conveniently: three from Kumba I and ten from Kumba II. Mambanda (Kumba III) was selected for piloting (Figure 1).

Third stage: Within each selected quarter, streets were randomly selected to guide participant recruitment. This selection incorporated local knowledge to ensure geographical and socioeconomic diversity.

Final stage: Participant recruitment used both purposive and random sampling at the household level, guided

by community health workers, and staff from the Cameroon Baptist Convention Health Services (CBCHS) HIV Free Project who conduct regular outreach programmes.

This community-engaged approach promoted cultural sensitivity, enhanced trust, and ensured inclusive representation. Prior to participation, individuals received verbal and written information about the study's aims and procedures, and written informed consent was obtained.

Sample size estimation

The sample size was estimated using the *CDC Epi Info™ 7.2.7.0 StatCalc* for cross-sectional studies, with outcome interest being “tested for HIV in the past 12 months.” Based on national data, recent HIV testing prevalence was approximated at 46.4% [19, 22]. Using standard parameters, 95% confidence level, 80% power, odds ratio of 2.0, and a 1:1 exposure ratio, the minimum sample size was calculated as 210 participants under simple random sampling.

To adjust for clustering due to the multi-stage sampling approach, the design effect ($Deff$) was estimated using the formula by Wan et al. (2021) [23]: $Deff = 1 + (m - 1) \times \rho$, where m is the average cluster size (10) and ρ is the intraclass correlation coefficient (0.05), yielding a $Deff$ of 1.45. The adjusted sample size became $210 \times 1.45 = 305$. To compensate for potential non-response, the sample was inflated by 15%, resulting in a final target of 351 participants.

Given the 1:4 sampling ratio between Kumba I and Kumba II municipalities, the sample was proportionally distributed: 71 participants in Kumba I and 280 in Kumba II.

Data collection tool and procedure

Data were collected using a structured, self-administered questionnaire (see *Supplementary File S1. Questionnaire*), developed in English and pre-tested for content validity and clarity. It comprised four sections: demographic data, knowledge about HIV testing, attitudes towards HIV testing, and practices of HIV testing.

The respondents were instructed to tick the appropriate box or fill in blank spaces as required. Trained research assistants supervised the process and ensured adherence to ethical and procedural protocols. Completion took about 20–30 min.

Questionnaire Validity and Reliability

The questionnaire was adapted from validated instruments used in previous studies [24, 25], with modifications tailored to reflect the sociocultural and epidemiological context of Kumba. A pilot test involving 36 participants in Mambanda (Kumba III) was conducted to assess for clarity, feasibility and reliability. Minor revisions were made to response options and item wording

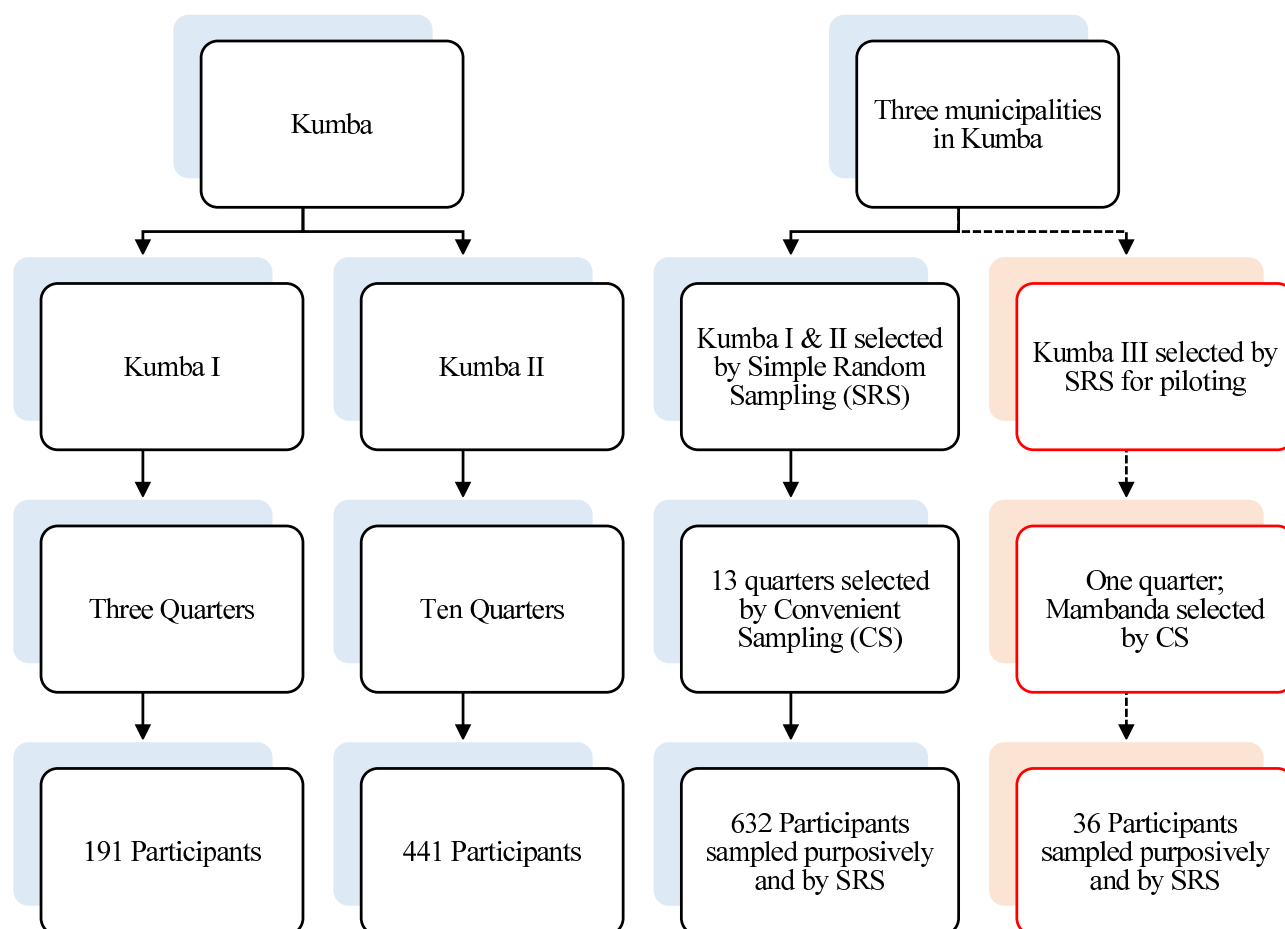


Fig. 1 Multistage sampling procedure for participant recruitment

for clarity. These participants were excluded from the main study analysis.

Internal reliability was conducted using Cronbach's alpha (α), a method appropriate for the sample size in accordance with psychometric recommendations [26]. Internal Consistency scores for each composite domain were: knowledge (13 items), attitudes (7 items), stigma perception (5 items), and testing practices (11 items). The internal consistency scores were as follows: knowledge ($\alpha=0.78$), attitudes ($\alpha=0.70$), perception of stigma ($\alpha=0.74$), and testing practices ($\alpha=0.81$). These values indicate acceptable to good internal consistency, in accordance with standard psychometric guidelines [26–28].

Ethical Considerations

Administrative authorisation was obtained from the Regional Delegate of Public Health for the Southwest Region of Cameroon (Ref: P42/MINSANTE/SWR/RDPH/CB.PT/529/430), in accordance with the Declaration of Helsinki [29, 30]. Additional administrative clearance was granted by local quarter heads. Written informed consent was obtained from all participants. This study followed the Strengthening the Reporting of

Observational Studies in Epidemiology (STROBE) guidelines for cross-sectional studies [31].

Variables and Measurement

Sociodemographic were collected using closed-ended items (Questions Q1–Q7). The main outcome variables were derived from four core domains: knowledge, attitudes, perceived stigma, and practices (KASP) towards HIV testing. Perceive stigma was assessed and analysed as a separate construct, distinct from attitudes, in order to evaluate its unique influence on testing behaviours.

For each domain, composite scores were calculated based on respondents' answers and categorised using percentiles into three levels: Low, Moderate, and High. Specifically, the categorisation was as follows:

Knowledge of HIV and HIV testing (K-score)

Assessed using 13 items (Q8–Q20) covering: Modes of HIV transmission, preventive strategies, HIV diagnostic methods, symptoms and treatment, and awareness of testing centres and information sources. The correct response for each item was “1” point, and incorrect

response or “Don’t know” was “0” points. The maximum score was 10, and categories scored as shown in Table 1.

Attitudes toward HIV testing (A-score)

This was measured using seven closed-ended items (Q21–Q27) that addressed the perceived importance of HIV testing, comfort with testing, perceived stigma, and support for mandatory testing. Although some items overlapped conceptually with stigma (e.g. Q22, Q24), these were analysed separately in the S-score to ensure analytical distinction. Each positive/favourable response(s) was scored as “1” point; each negative/unfavourable response was scored “0” points. The total score was converted to a percentage and grouped into three categories [12, 32]: negative ($\leq 27.8\%$), neutral ($> 27.8\%$ and $\leq 38.9\%$), and positive ($> 38.9\%$), as shown in Table 1. The “neutral” category represents participants whose attitudes neither

consistently aligned with positive nor negative perspectives across the seven items.

Perception of stigma (S-score)

Perception of stigma related to HIV testing was evaluated with responses from four items (Q22, Q24–27) that assessed fears of societal judgment, shame, and rejection. Responses indicating stigma were scored 1 point, while non-stigma responses scored 0 points. The total score was then categorised using percentiles: Low (0.0–33.3%), Moderate (33.4–66.6%) and High (66.7–100%).

Practices related to HIV testing (P-score)

Practices related to HIV testing were assessed using 11 items (Q28–Q38) including: History of HIV testing (self and partner), frequency and timing of tests, Use of HIV support systems, and willingness to test in the future. Scores were assigned based on frequency and positivity of responses, with a maximum score of eight.

These four domains were analysed both as categorical outcomes (e.g., high/moderate/low) and as independent variables.

Table 1 Dependent variables, items, scoring, and interpretation

Domain	Item Question N°	Description	Scoring	Score range	Interpretation categories
Knowledge of HIV and HIV testing (K-score)	Q8–Q20	Assesses knowledge of HIV transmission, prevention, diagnosis, symptoms, and treatment	Correct = 1 point Incorrect/Don't know = 0 points	0–32	Low (0–39), Moderate (40–69), High (70–100) [32, 33]
Attitudes toward HIV testing (A-score)	Q21–Q27	Assesses importance, comfort, stigma perception, and beliefs around HIV testing	Positive response = 1 point Negative = 0 points	0–16	Negative (≤ 27.8), Neutral (> 27.8 and ≤ 38.9), Positive (> 38.9) [12, 32]
Perception of stigma (S-score)	(Q22, Q24–27)	Evaluates the respondent's perception of social stigma associated with HIV testing, including fear of judgment, shame, and rejection	Stigma-related response = 1 point Non-stigma response = 0 points	0–12	Low (0.0–25%), Moderate (25–50%), High (50–100%)
Practices related to HIV testing (P-score)	Q28–Q38	Evaluates testing history, frequency, partner testing, and use of support services	Based on frequency and pro-activity; points per question	0–33	Poor (0–39), Moderate (40–69), Good (70–100) [16, 34]

Note: Percentile-based cut-offs were applied to the three-point ordinal scale scores to classify composite scores into low, moderate, and high categories or negative, neutral, and positive categories, depending on the domain. References [12, 16, 32–34] provide further validation of scoring frameworks

Perception of stigma (S-score)

Data Management and Analysis

All data were entered into Microsoft Excel and analyses were conducted using SPSS version 25 (IBM Corp., Armonk, NY, USA). A consistency check was performed between two questionnaire items Q8 (knowledge domain) and Q28 (practice domain), both of which assessed whether the respondent had ever been tested for HIV. Of the 645 respondents, 67 (10.4%) provided discordant answers. Question 28 (Q28), situated within the practice section and aligned with behavioural measurement, was designated as the primary outcome variable for HIV testing history. Fifty-four discordant cases were retained in the primary analysis but flagged for sensitivity analysis and addressed in the limitations.

To evaluate the robustness of findings, a sensitivity analysis was conducted by excluding the 67 discordant responses. Key bivariate and multivariable regression models were re-estimated. The direction, magnitude, and statistical significance of associations remained largely unchanged, indicating that the primary findings were stable and not materially affected by potential internal reporting inconsistencies.

Descriptive statistics were calculated for all variables. Categorical variables were summarised using frequencies and percentages, while continuous variables were described using means and standard deviations. Bivariate associations were tested using Pearson's Chi-squared (χ^2) test. Variables with significant p-values from univariate analysis were considered for inclusion in the multivariable logistic regression model to identify independent

predictors of HIV testing. The regression model estimated odds ratios (ORs) with 95% confidence intervals (CIs) for each predictor variable, adjusting for potential confounders. The final model aimed to assess the strength of associations between the predictors and HIV testing behaviour, controlling for other factors. With statistical significance was set at $p < 0.05$.

To assess multicollinearity among independent variables in the multivariable logistic regression models, we examined tolerance and variance inflation factor (VIF) values. All tolerance values were above 0.7 and VIF values ranged between 1.05 and 1.38 across models for knowledge, attitudes, and practices (Supplementary Table 1),

Table 2 Demographic characteristics of study participants ($n = 632$)

Characteristic	Subclass	Frequency (%)	Ever tested for HIV ($n = 476$)	χ^2 (p-value)
Age (years)	18–30	559 (88.4)	412 (86.6)	6.776 (0.009)
	31–35	73 (11.6)	64 (13.4)	
	Mean ($\bar{x} \pm SD$)	23.5 $\pm$ 4.9	23.7 $\pm$ 5.1	
Gender	Male	223 (35.3)	139 (29.2)	31.251 (< 0.001)
	Female	409 (64.7)	337 (70.8)	
Marital Status	Single	441 (69.8)	319 (67.0)	6.974 (0.008)
	Married	191 (30.2)	157 (33.0)	
Educational Status	Tertiary	145 (22.9)	116 (24.4)	2.244 (0.326)
	Secondary	358 (56.6)	264 (55.5)	
	Primary	129 (20.4)	96 (20.2)	
Current occupation	Student	443 (70.1)	333 (70.0)	7.557 (0.056)
	Employed	75 (11.9)	63 (13.2)	
	Self-employed	91 (14.4)	67 (14.1)	
	Others	23 (3.6)	13 (2.7)	
Religion	Christianity	540 (85.4)	399 (83.8)	4.112 (0.128)
	Islam	65 (10.3)	54 (11.3)	
	African Traditional Religion	27 (4.3)	23 (4.8)	
Number of sex partners	None	256 (40.5)	175 (36.8)	11.834 (0.003)
	One	261 (41.3)	212 (44.5)	
	Two-Four	115 (18.2)	89 (18.7)	
	Mean ($\bar{x} \pm SD$)	0.8 $\pm$ 0.9	0.9 $\pm$ 0.9	
Municipality	Kumba I	191 (30.2)	149 (31.3)	1.069 (0.301)
	Kumba II	441 (69.8)	327 (68.7)	

Note: Values are presented as number (percentage), Bold numbers indicate significant p-values for Pearson Chi Square (χ^2)

SD Standard Deviation, $\bar{x}$, mean

indicating no concerns of multicollinearity according to standard thresholds (tolerance < 0.1 or VIF > 10).

Results

A total of 700 questionnaires were distributed across the two municipalities: 200 in Kumba I and 500 in Kumba II. Of these, 181 (90.5%) from Kumba I and 464 (92.8%) from Kumba II were deemed eligible and included in the analysis, yielding a preliminary total of 645 (92.1%). However, 67 responses showed discrepancies between Q8 and Q28. Thirteen of these were excluded, resulting in a final analytical sample of 632.

Participant Demographics

Table 2 presents the demographic characteristics of the 632 study participants, stratified by self-reported history of HIV testing. The majority of participants were aged between 18 and 30 years ($\bar{x} \pm SD$; 23.7 $\pm$ 5.1 years) (88.4%), and students (70.1%).

A statistically significant association with HIV testing history was observed for age groups ($\chi^2 = 6.776$, $p = 0.009$), gender ($\chi^2 = 31.251$, $p < 0.001$), marital status ($\chi^2 = 6.974$, $p = 0.008$), and number of sexual partners ($\chi^2 = 11.834$, $p = 0.003$).

Knowledge about testing, attitudes and perceptions of stigma towards HIV testing, and practices related to HIV testing

Knowledge and awareness of HIV and testing

The overall level of knowledge about HIV testing was moderate, with a mean score of 49.8 $\pm$ 9.3. Most participants 528 (83.5%) had moderate level, 81 (12.8%) had low knowledge, and only 33 (3.6%) had high knowledge (Supplementary Fig. 1).

While general awareness of HIV exists, significant misinformation about prevention and treatment persists, requiring targeted educational interventions.

Knowledge and testing practices

Four hundred and sixty (72.8%) participants had been tested for HIV at least once, while 172 (27.2%) had never tested. Among those who had tested, over half (58.9%) reported testing every three months (Table 3).

Understanding of HIV testing methods

Almost all participants, 622 (98.4%), identified blood as the main sample used for HIV testing. A small proportion also mentioned urine, 48 (7.6%), and saliva, 38 (6.0%). Participants reported “saliva”, likely referring to the oral fluid used in HIV testing. This term is retained for authenticity but clarified here to reflect clinical accuracy.

Table 3 Perceived knowledge and awareness of HIV and testing ($n=632$)

Characteristic	Subclass	Frequency (n)	Percentage	
Tested for HIV	Yes	460	72.8	
	No	172	27.2	
Knowledge of when HIV testing should be done (Q9)	Not at all	75	11.9	
	Whenever possible	134	21.2	
	Once a year	51	8.1	
	Every three months	372	58.9	
Samples used for testing†	Blood	622	98.4	
	Saliva	38	6.0	
	Urine	48	7.6	
	Blood, Saliva, and urine	38	6.0	
How can someone determine whether they have HIV/AIDS?	Consult a medical doctor	142	22.5	
	Hospital laboratory results	487	77.1	
	Health personnel	13	2.1	
Modes of HIV transmission†	Sexual intercourse	505	79.9	
	Blood transfusion	282	44.6	
	Mother-to-child (birth)	90	14.2	
	Sexual intercourse, transfusion, birth	73	11.6	
Prevention of HIV transmission†	Abstinence from sex	186	29.4	
	Fidelity	152	24.1	
	Use of condoms	348	55.1	
	Use of contraceptives	425	62.7	
	Avoid sharing of cutlery with infected persons	33	5.2	
Treatment and curability knowledge	HIV can be treated	466	73.7	
	HIV can be cured	143	22.6	
	HIV can be treated but cannot be cured	310	49.1	
Understanding causes of HIV†	Virus	573	90.7	
	Sin	34	5.4	
	Curse	18	2.8	
	Bad spirit	7	1.1	
Common symptoms of HIV†	Fever	295	46.7	
	Fatigue	189	29.9	
	Swellings (groin or armpits)	214	33.9	
	Night sweats	49	7.8	
Treatment mechanism	Stops the HIV virus from reproducing	476	75.3	
	Cures AIDS completely	156	24.7	
Sources of HIV information†	Mass media	Television	118	18.7
		Radio	150	23.7
		Both TV and radio	93	14.7
	Print media	Newspapers	119	18.8
	Interpersonal sources	Healthcare officials	441	69.8

Note: (1) Q9 perceived or recommended testing frequency (knowledge domain), (2) Saliva, as used here, reflects participant-reported terminology and likely refers to oral fluid used in HIV oral testing

[†]Characteristic has multiple responses. Frequencies and percentages reflect the number of participants who selected each barrier; totals exceed 100% due to multiple selections

Four hundred and eighty-seven (77.1%) of the participants acknowledged that one could know if they are infected with HIV from hospital laboratory results, while 142 (22.5%) acknowledged the medical doctor.

Awareness of HIV transmission

Approximately four-fifths (79.9%) of the participants correctly stated that HIV can be transmitted through unprotected sex, 282 (44.6%) said it could be transmitted

through blood transfusion, and 90 (14.2%) mentioned mother-to-child transmission.

Knowledge on prevention

Four hundred and twenty-five (62.7%) of the participants wrongly asserted that HIV could be prevented with contraceptives, while 348 (55.1%) and 186 (29.4%) correctly cited condoms and abstinence, respectively. A few

participants 33 (5.2%) incorrectly thought that avoiding sharing cutlery with infected individuals, could also prevent HIV.

Understanding of treatment and cure

About three-quarters [466 (73.7%)] of the participants believed that HIV can be treated. However, 143 (22.6%) wrongly believed that it could be cured, while 310 (49.1%) correctly stated that it could be treated but not cured.

Concerning how HIV treatment works, 476 (75.3%) of the participants knew that the antiretroviral therapy prevents the virus from reproducing, yet 156 (24.7%) wrongly believed that treatment comple

Beliefs about the cause of HIV

Five hundred and seventy-three (90.7%) of the participants correctly stated that HIV is caused by a virus, but a few still linked it to sin 34 (5.4%), curses 18 (2.8%), and bad spirits 7 (1.1%), indicating the persistence of traditional beliefs.

Table 4 Attitudes and perceptions to stigma related to HIV testing among youths in Kumba ($n=632$)

Characteristic	Subclass	Frequency (n)	Percentage
Perceived benefits of HIV testing	Very important	204	32.3
	Somewhat important	250	39.6
	Not important	85	13.4
	I don't know	93	14.7
Perception that people with HIV are treated unfairly	Yes	251	39.7
Comfort with HIV testing	Very comfortable	198	31.3
	Moderately comfortable	116	18.4
	Very uncomfortable	318	50.3
Barriers discouraging HIV testing [†]	Lack of awareness	138	21.8
	Stigma	107	16.9
	Symptomless	57	9.0
	Confidentiality	32	5.1
	Culture	15	2.4
	Fear of positive results	325	51.4
	No trust in health-care services	19	3.0
Embarrassed if friends knew about the HIV test	Yes	256	40.5
Belief HIV testing should be compulsory for youth	Yes	454	71.8
Belief that people in Kumba fear HIV test results	Yes	446	70.6

Frequencies and percentages reflect the number of participants who selected each barrier, totals exceed 100% due to multiple selections

[†]Characteristic has multiple responses

Recognition of symptoms

Of the common symptoms enumerated, fever was the most recognised, by 295 (46.7%), followed by swollen lymph nodes in the groin or armpits 214 (33.9%) as presented in Table 3.

Sources of HIV information

Interpersonal sources: healthcare workers were the most common source of information 441 (69.8%). Participants also received HIV information from the mass media; radio 150 (23.7%), and television 118 (18.7%).

Attitudes and perceptions of stigma towards HIV testing

Participants' overall attitude towards HIV testing was broadly neutral with a mean score of 68.7 ± 10.2 . This value falls within the upper range of the "neutral" category (27.9–38.9%), and just below the threshold for a "positive" classification (> 38.9%). Although only 128 (20.3%) of the participants were formally classified as having a positive attitude, a substantial number had scores that were closer to the boundary, which helps explain the relatively high mean despite the low proportion of positive classifications. Specifically, 282 (44.6%) participants fell into the neutral attitude, indicating ambivalence or a balance of favourable and unfavourable views. In contrast, 222 (35.1%) held a negative attitude, while only 128 (20.3%) were classified as having a positive attitude towards HIV testing (Supplementary Fig. 1).

Perceived benefits of HIV testing

Two hundred and four (32.3%) participants considered HIV testing to be very important, while 250 (39.6%) described it as 'somewhat important' (Table 4). These findings suggest a favourable perception of HIV testing among youths, although 93 (14.7%) remain uncertain or dismissive of its value.

A large majority of the participants, 454 (71.8%), supported systematic approaches to testing, agreeing that HIV testing should be made compulsory. This study suggests a strong endorsement of structured HIV prevention strategies among the youth population.

Personal experiences with testing

Only 198 (31.3%) of the study participants felt 'very comfortable' undergoing HIV testing, while 116 (18.4%) described themselves as 'moderately comfortable'. More than half (50.3%) expressed feeling 'very uncomfortable', suggesting significant psychological or contextual barriers to testing.

The main barriers discouraging testing included 'fear of a positive result', 325 (51.4%), 'lack of awareness' 138 (21.8%), and 'stigma' 107 (16.9%).

Despite these barriers, some participants appeared to understand the social implications of HIV testing: 256

(40.5%) admitted they would feel embarrassed if their friends knew they had taken an HIV test.

Perceptions of stigma towards HIV testing

A three-point ordinal stigma scale was used to classify participants into levels of perceived stigma (Supplementary File S2. Supplementary Tables). Stigma levels were categorised as low 382 (60.4%), moderate 233 (36.9%), and high 17 (2.7%). The mean stigma score was 27.7 ± 9.1 , indicating low-to-moderate stigma scores, and a non-negligible subgroup experienced high levels of perceived stigma.

Perceptions of the stigma surrounding HIV and HIV testing were prevalent. A significant number of participants 251 (39.7%) believed that PLWHA are treated unfairly in the community. Further, 446 (70.6%) participants admitted that individuals usually fear receiving their HIV test results, further emphasising anxiety and potential discrimination.

The findings reveal that despite supportive attitudes toward testing policies, personal stigma and emotional discomfort remain substantial barriers.

Practices related to HIV testing

Despite a reasonably high willingness to test 491 (77.7%), overall practice levels towards HIV testing were generally poor [average score of 30.7 ± 15.7]. Of the 632 participants, 450 (71.2%) had a poor practice score, 179 (28.3%) had a moderate score, and only 3 (0.5%) demonstrated good practice (Supplementary File S2. Supplementary Tables).

Table 5 presents the HIV testing practices based on key themes: testing history, partner testing behaviours, testing frequency, support systems, HIV testing barriers, and willingness to test.

Testing history

Approximately three-quarters (75.3%) of the participants, reported that they had been tested for HIV at least once before. Non-tested participants were about one-fifth (21.0%), while the majority 375 (59.3%) had yearly testing. Very few participants admitted testing for HIV more often, with 36 (5.7%) reporting that they test three times a year and 88 (13.3%) saying they test four times a year. In the past year, 300 (47.5%) had been tested within 1–3 months ago, 50 (7.9%) within 4–6 months ago, while 207 (32.8%) had not been tested within the past year. The average reported testing frequency was 2.4 ± 2.9 months.

Partner testing

Of the 558 participants who answered the question on partner testing, 393 (70.4%) reported that their partner had never been tested. Conversely, 150 (26.9%) participants stated that their partner had been tested within

the past 1–3 months. A smaller proportion reported that their partners had been tested within the last 4–6 months and 7–9 months, with 9 (1.6%) and 6 (1.1%) participants, respectively. Regarding the frequency of partner testing in the past year, 323 out of 479 (51.1%) indicated that their partner had not been tested, 96 (15.2%) reported 1–3 times and 15 (2.4%) stated 4–6 times.

Frequency of HIV testing

A total of 169 out of the 616 participants (27.4%) who responded to this question reported that they had not been tested for HIV during the past year. In contrast, 263 (42.6%) participants reported that they were tested once, 132 (21.4%) were tested 2–3 times, and 52 (8.4%) were tested 4–5 times (Table 5). These findings indicate that although a majority, 447 (72.6%), of participants had at least one HIV test in the past year, only 184 (29.8%) were tested more than once. This highlights the need to strengthen public health strategies that encourage regular and repeated HIV testing among the youth.

Support systems use and awareness

Most [338 (53.5%)] of the participants of the survey were able to identify and acknowledge the existence of support systems. Of these, the major cited support system was school programmes by 248 (39.2%) along with community outreach programmes 132 (20.9%). A hundred and nine (17.2%) of the participants mentioned youth health services, while 55 (8.7%) mentioned peer support or peer education (Table 5).

Barriers to testing

The most common barriers reported were 265 (41.9%) fear of positive results, 101 (16.0%) lack of awareness or education, 76 (12.0%) lack of HIV symptoms, and 76 (12.0%) concerns over confidentiality. Other barriers included fear of stigma, no access to testing, and distrust in health care services (Table 5).

Willingness to test now

When asked about their current willingness to test, 491 (77.7%) participants responded positively.

Socioeconomic factors influencing HIV testing

Supplementary Table 2, shows associations between various sociodemographic factors and the three domains of HIV testing: knowledge, attitudes, and practices.

Participants with no sexual partners were thrice more likely to have good HIV knowledge than those with two to four partners (OR 3.0, 95% CI: 1.5–5.8, $p=0.001$). Those with neutral (OR 3.0, 95% CI: 1.5–5.7, $p=0.001$) and negative attitudes (OR 2.1, 95% CI: 1.0–4.1, $p=0.038$) had higher odds of good knowledge. Low perceived stigma significantly decreased the likelihood of poor knowledge

Table 5 Practices towards HIV testing ($n=632$)

Theme	Characteristic	Subclass	Frequency	Percentage	
Testing history	Ever been tested	Yes	476	75.3	
		Not tested	133	21.0	
	Frequency of HIV testing	Once a year	375	59.3	
		Three times yearly	36	5.7	
		Four times yearly	88	13.3	
		When last tested for HIV in last year	Not tested	207	32.8
		1–3 months ago	300	47.5	
		4–6 months ago	50	7.9	
		7–9 months ago	46	7.3	
		10–12 months ago	29	4.6	
		Mean ($\bar{x} \pm \text{SD}$)	2.4 $\pm$ 2.9		
Partner testing	^a When last tested for HIV by partner	Not tested	393	70.4	
		1–3 months ago	150	26.9	
		4–6 months ago	9	1.6	
		7–9 months ago	6	1.1	
	^b Frequency of partner HIV testing	Not time	323	51.1	
		1–3 times	96	15.2	
HIV testing frequency	^c Number of times tested last year	4–6 times	15	2.4	
		Not time	169	27.4	
		One time	263	42.6	
		2–3 times	132	21.4	
	4–5 times	52	8.4		
Support Systems Use and Awareness	Are there HIV support systems	Yes	338	53.5	
	Available support systems	Peer Support/Education	55	8.7	
		School programmes	248	39.2	
		Youth Health Services	109	17.2	
		Community outreach programmes	132	20.9	
		Healthcare providers	48	7.6	
		Parent or family	17	2.7	
		No trust for the HCS	26	4.1	
Barriers to HIV testing	Barriers ^d	Fear of stigma	69	10.9	
		Lack of awareness/education	101	16.0	
		Fear of Positive results	265	41.9	
		Lack of symptoms	76	12.0	
		Confidentiality	76	12.0	
		No access to testing	26	4.1	
		No reason	44	7.0	
		Willingness to test now	Yes	491	77.7
		Willingness to test now	Yes	491	77.7

Frequencies and percentages reflect the number of participants who selected each barrier; totals exceed 100% due to multiple selections.

^a558 participants in this category, ^b479 participants in this category, ^c616 participants in this category, ^dCharacteristic has multiple responses.

(OR 0.4, 95% CI: 0.2–0.6, $p=0.001$). Good HIV-related practices were also associated with lower odds of poor knowledge (OR 0.1, 95% CI: 0.01–0.9, $p=0.044$).

High knowledge strongly predicted positive attitudes (OR 5.2, 95% CI: 2.2–12.3, $p<0.001$). Participants with no sexual partners (OR 3.2, 95% CI: 1.8–5.5, $p<0.001$) or one partner (OR 1.9, 95% CI: 1.1–3.2, $p=0.015$) were more likely to report favourable attitudes. Low perceived stigma was protective (OR 0.1, 95% CI: 0.1–0.2, $p<0.001$), while high stigma increased the odds of negative attitudes (OR 2.7, 95% CI: 1.4–21.9, $p=0.002$).

No sexual partners remained the strongest predictor of good practices (OR 7.6, 95% CI: 4.1–14.3, $p<0.001$), followed by having one partner (OR 2.2, 95% CI: 1.3–3.8, $p=0.003$). Low stigma increased the odds of good practices (OR 2.1, 95% CI: 1.3–3.2, $p=0.001$), whereas high stigma reduced them (OR 0.3, 95% CI: 0.1–0.9, $p=0.042$).

Discussions, recommendations and conclusions

Summary of key findings

This study evaluated HIV knowledge, attitudes, stigma perceptions, testing practices, and their socioeconomic determinants among youths aged 18–53 years in Kumba,

Cameroonian. Although a majority, 460 (72.8%) of participants reported ever testing for HIV, recent and routine testing remained low, and stigma was a persistent deterrent. The knowledge of HIV was generally moderate, although misconceptions about transmission and prevention persisted, particularly around where to test and the availability of treatment. Attitudes towards HIV testing were mixed, with only 128 (20.3%) participants expressing positive views. Majority of the participants held neutral or uncertain attitudes. Sociodemographic factors, especially the number of sexual partners and level of perceived stigma, significantly influenced HIV-related knowledge, attitudes, and testing practices.

Discussions

Knowledge about HIV and testing

Participants demonstrated average awareness of HIV risk factors and prevention, but specific knowledge gaps persisted particularly regarding the testing frequency, modes of transmission as well as prevention. This pattern aligns with previous studies in Nigerian and South Africa which found that while awareness of HIV was high, specific knowledge about testing was limited [35, 36]. In our study, only 33 (3.6%) of the participants demonstrated high HIV testing knowledge, which was far lower than the 44.7% reported in South Africa [25], and 61.6% in Takoradi metropolis, Ghana [37].

Our moderate knowledge score 528 (83.5%) compares favourably to the 63.6% reported in the Ho municipality in Ghana [38]. In our study, almost all participants could name at least one transmission route, consistent with findings by [39], who also demonstrated that increased HIV knowledge correlates with testing uptake. The implication is that while general awareness is high, targeted education to address practical knowledge is still necessary.

Attitudes and perceptions of stigma towards HIV testing

Attitudes towards HIV testing

Overall attitudes toward HIV testing were generally favourable with a mean score of 68.7%, but only 128 (20.3%) participants exhibited a distinctively positive attitude. This was lower than the 76% reported by [12] in Fako, but higher than the 26.7% reported among slum dwellers in Bangladesh [40], and 13.2% among pharmacy students in Saudi Arabia [24]. The 222 (35.1%) negative attitudes, was less than the 50% reported among Cameroonian secondary school students [7].

Age appeared to influence attitudes, with younger respondents more likely to perceive testing as a responsible and normalised health behaviour. These views are echoed by Katirayi et al. (2023), who noted that men of aged 21–30 in Cameroon increasingly regarded HIV testing as routine. In contrast, older participants expressed

concerns that testing could cause workplace disruption or social exclusion [41], suggesting that fear of social consequences may persist among older youth subgroups. Thus, while testing is increasingly accepted among adolescents and young adults, practical and cultural apprehensions persist, particularly among the older segment of “youth”.

Perceptions of stigma towards HIV testing

Despite moderate knowledge and favourable attitudes, stigma-related concerns were evident. Participants reported fears of gossip, discrimination, and social rejection if seen testing or receiving a positive result. Many participants reported fears of discrimination, gossip, and social rejection if seen testing or receiving a positive result. These concerns align with broader findings in sub-Saharan Africa [42, 43], where internalised and anticipated stigma reduce voluntary testing rates.

Sabo et al. (2024) [44] found that discriminatory attitudes towards people living with HIV were inversely associated with testing uptake, consistent with our findings. Addressing stigma through peer education, safe spaces, and community normalisation campaigns remains essential to bridge the gap between intention and testing behaviour.

Practices related to HIV testing

Despite the fact that 460 (72.8%) had previously tested for HIV as well as generally favourable attitudes, only a minority reported testing in the past three months, and partner testing was infrequent. These behaviours suggests that testing is often opportunistic rather than routine, often linked to campaigns, school events or emergencies. Similar trends were reported in Cameroon by [45], where only 16.1% had tested in the past three months, and by [19], who found that the national testing rates were 47.8% for women and 45.0% for men.

Persistent stigma, lack of youth-centred services, and fear of positive results may explain the low recent testing uptake, even among those who had tested at least once. Interventions that integrate testing into routine youth health services and reduce access barriers are urgently needed.

Socioeconomic factors influencing HIV testing

Participants with no sexual partners demonstrated significantly higher odds of good HIV testing knowledge (OR 2.98, 95% CI: 1.76–5.04, $p < 0.001$), positive attitudes (OR 3.15, 95% CI: 1.89–5.23, $p < 0.001$), and good practices (OR 7.62, 95% CI: 4.05–14.32, $p < 0.001$). This is consistent with previous research suggesting that individuals not engaged in sexual activity may be more receptive to HIV education and less influenced by stigma [46]. Neutral and negative attitudes were also significantly

associated with higher knowledge scores (neutral: OR 2.96, 95% CI: 1.72–5.08, $p < 0.001$; negative: OR 2.08, 95% CI: 1.18–3.67, $p = 0.011$), possibly reflecting an ongoing transition in behavioural understanding, as similarly noted by [47]. Lower perceived stigma was significantly associated with improved knowledge and testing behaviours, corroborating findings by [10, 11].

In contrast to previous studies [14, 19], we found no significant associations between gender, education, or religion and testing behaviour, highlighting the complex interplay between individual, social and structural factors in this context. These findings suggest targeted stigma-reduction and sexuality-sensitive interventions could enhance HIV testing engagement in this population, consistent with [48] guidelines promoting tailored youth strategies.

Public health implications

The findings have clear implications for multiple stakeholders of HIV prevention in Cameroonian.

- For primary beneficiaries, namely youths aged 18–35 years, the findings reinforce the need for stigma-reduction strategies and culturally adapted health messaging to increase HIV testing. School-based education should not only address knowledge gaps but also dispel common myths that fuel testing avoidance. Engaging peer leaders and involving families in awareness campaigns may help reduce fears of discrimination. To encourage uptake, testing services should be youth-friendly, offering flexible hours, mobile clinics, and, where feasible, technology-based reminders or support systems.
- Among secondary beneficiaries such as health professionals, teachers, and community workers, training programmes should focus on improving youth-friendly service delivery, stigma-sensitive counselling, and effective community engagement. These stakeholders play a critical role in shaping the testing environment and should be well equipped to support young people who fear judgement or social consequences, especially in school and workplace settings.
- For tertiary stakeholders, including policymakers, NGOs, and global health actors, efforts should prioritise policies and funding that support decentralised HIV testing services, mobile outreach, and the integration of youth-focused HIV education into broader health systems. While digital platforms hold promise for expanding access and engagement, their implementation should be context-specific and guided by further evidence from local settings.

Limitations and strengths of the study

Limitations of the study

The cross-sectional design of this study means we cannot determine the direction of influence between HIV testing and the knowledge or attitudes that came before it. Self-reported data may be affected by social desirability or recall bias, potentially causing some participants to overstate positive behaviours like HIV testing.

A consistency check revealed that 10.4% of respondents gave conflicting answers about their testing history. Although these discrepancies were addressed by prioritising one question and conducting sensitivity analyses, which showed stable results, the inconsistencies highlight possible reporting errors.

The questionnaire, while structured to provide clear responses, may not fully capture beliefs specific to the local culture or region. Additionally, although the sample included youths from different areas, out-of-school youths or those in very remote rural locations might be underrepresented, limiting the generalisability of the findings.

Finally, ongoing changes in HIV services, such as the introduction of self-testing, may mean some results become less relevant over time.

Strengths of the study

Despite the limitations, the study also had notable strengths. We used a large, representative dataset and validated tools, which strengthens the relevance of our findings. The broad coverage across municipalities, and socioeconomic groups suggests the results reflect real population patterns. Standardised questions also allow comparison with other studies. We focused specifically on youth aged 18–35, which is broad. By examining knowledge, attitudes, stigma, and practices together, we gained a fuller understanding of what drives HIV testing. These strengths support the view that the findings reflect real challenges faced by youth.

Suggestions for future studies

Future research should build on these insights. Longitudinal studies would help clarify whether changes in knowledge or stigma lead to later testing. Qualitative work, like focus groups or interviews, could explore why some youths hesitate to test and how cultural beliefs influence this. Given interest in digital tools, pilot projects using mobile apps for self-testing reminders or counselling should be developed and evaluated. Future studies should also assess the impact of stigma-reduction campaigns by comparing testing rates before and after such interventions. It is important to include harder-to-reach groups internally displaced victims of the ongoing crisis, or young sex workers. Strong partnerships with community health workers will be key to reaching them.

Conclusions

Summarily, this study shows a mixed picture: Cameroonian youth have good basic knowledge of HIV and mostly support testing, but key gaps remain. Misconceptions and fear of stigma still stop some from testing. Socioeconomic factors like education and number of sex partners affects who gets tested. To make progress, interventions must focus on youth, be culturally appropriate, and address multiple barriers; through better sex education, stigma reduction, and easier access to testing. Using mass media and mobile tools may help reach this group. Closing knowledge gaps, tackling stigma, and reducing inequality are key to improving youth HIV testing rates.

Abbreviations

AIDS	Acquired immunodeficiency syndrome
HIV	Human immunodeficiency virus
KAP	Knowledge, Attitudes, Practices
PLWHIV	People living with HIV
WHO	World Health Organisation

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12889-025-24229-3>.

Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Supplementary Material 4.

Supplementary Material 5.

Acknowledgements

The authors would like to thank all study participants for their time and willingness to contribute to the research, particularly those who consented to HIV screening. We are deeply grateful to the HIV Free staff at Centre Médical d'Arrondissement (CMA) Ntam, the Catholic Integrated Health Centre (CIHC) Kumba, and the Kumba Regional Hospital Annex (KRHA) for their invaluable support during fieldwork. We also extend our sincere appreciation to the laboratories of CMA Ntam, CIHC Kumba, and KRHA for confirming reactive HIV results from community testing, as well as conducting verification testing, counselling and linking to HIV care and ARTs. Our thanks also go to the staff and students of the RAPHA Higher Institute of Health and Biomedical Science in Kumba for their contributions to the study.

Authors' contributions

Conceptualisation: MCFN, FNC, CAT. Data curation and formal analyses: MCFN, FNC, DZT, YMC, JPK, EM. Investigation: MCFN, DZT, SAK. Methodology: MCFN, FNC, CAT, FNN, JPK, EM, ANT. Project administration: MCFN, FNC, CAT, FNN, EMN. Resources: MCFN, FNC, EMN. Supervision: FNC, CAT, FNN, YMC, EMN, ANT. Validation: MCFN, FNC, DZT, SAK, CAT, FNN, YMC, JPK, EM, EMN, ANT. Writing—original draft: MCFN, FNC, CAT. Writing—review and editing: MCFN, FNC, DZT, CAT.

Funding

There was no funding for this study.

Data availability

All relevant data that support the conclusion of this study are included in the article.

Declarations

Ethics approval and consent to participate

This study was conducted in strict adherence to the principles outlined in the Declaration of Helsinki and received ethical approval from the Regional Delegation of Public Health for the Southwest Region, under approval number P42/MINSANTE/SWR/RDPH/CB.PT/529/430. The Quarter Heads of the selected quarters were approached with the objectives of the study, to obtain verbal administrative clearance. Informed consent has been obtained from all participants prior to their inclusion in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹RAPHA Higher Institute of Health and Biomedical Sciences, Kumba, Cameroon

²Cameroon Baptist Convention Health Services - HIV free/Strengthening Public Health Laboratory Systems, Douala, Cameroon

³Department of Nursing, Catholic University Bamenda - Kumba Campus, Kumba, Cameroon

⁴Department of Microbiology and Parasitology, University of Bamenda, Bamenda, Cameroon

⁵College of Medicine and Health Sciences, School of Public Health, University of Rwanda, Kigali, Rwanda

⁶Department of Obstetrics and Gynaecology, University of Buea, Buea, Cameroon

Received: 21 May 2025 / Accepted: 29 July 2025

Published online: 24 September 2025

References

- UNAIDS. Global HIV & AIDS statistics — Fact sheet [Internet]. 2024 [cited 2025 Jan 22]. Available from: <https://www.unaids.org/en/resources/fact-sheet>
- UNAIDS. 2024 global AIDS report — The Urgency of Now: AIDS at a Crossroads | UNAIDS [Internet]. 2025 [cited 2025 Apr 28]. Available from: <https://www.unaids.org/en/resources/documents/2024/global-aids-update-2024>
- WHO. HIV data and statistics [Internet]. 2024 [cited 2025 Jan 22]. Available from: <https://www.who.int/teams/global-hiv-hepatitis-and-stis-programmes/hiv/strategic-information/hiv-data-and-statistics>
- WHO. HIV and AIDS [Internet]. HIV/AIDS fact sheet. 2020 [cited 2025 Mar 18]. Available from: <https://www.who.int/news-room/fact-sheets/detail/hiv-aids>
- Zhang J, Ma B, Han X, Ding S, Li Y. Global, regional, and national burdens of HIV and other sexually transmitted infections in adolescents and young adults aged 10–24 years from 1990 to 2019: a trend analysis based on the Global Burden of Disease Study 2019. *The Lancet Child & Adolescent Health* [Internet]. 2022 Nov 1 [cited 2025 May 20];6(11):763–76. Available from: [https://www.thelancet.com/journals/lanchi/article/PIIS2352-4642\(22\)00219-X/abstract](https://www.thelancet.com/journals/lanchi/article/PIIS2352-4642(22)00219-X/abstract)
- UNICEF. Health & HIV | UNICEF Cameroon [Internet]. 2024 [cited 2025 May 10]. Available from: <https://www.unicef.org/cameroon/what-we-do/health-hiv>
- Nubed CK, Akoachere JFTK. Knowledge, attitudes and practices regarding HIV/AIDS among senior secondary school students in Fako Division, South West Region, Cameroon. *BMC Public Health* [Internet]. 2016 Aug 22 [cited 2025 Jan 27];16(1):847. Available from: <https://doi.org/10.1186/s12889-016-3516-9>
- Sanou SM, Fokam JM, Mabvouna BR, Guetiya WR, Sali BBA, Teikeu TVV, Nafack SS, Panà A, Colizzi V, Russo G. Knowledge, attitudes and practices on HIV / AIDS in the south region of Cameroon: case of the town of Kribi. *Ig Sanita Pubbl*. 2014;70(4):381–92.
- Sobze Sanou M, Fokam J, Guetiya Wadoun R, Russo G, Onohiol JF, Djeunang DB, Nkamedjie PP, Zambou NF, Zefack Y, Galletta A, Arduini M, Franca E, Panà A, Colizzi V. Condom perception and prevention of HIV/AIDS infection in Cameroon: appraisal of knowledge, attitudes and practices among level one students of the University of Dschang. *Ig Sanita Pubbl*. 2013;69(2):183–94.

10. Egbe TO, Nge CA, Ngouekam H, Asonganyi E, Nsagha DS. Stigmatization among People Living with HIV/AIDS at the Kumba Health District, Cameroon. *J Int Assoc Provid AIDS Care*. 2020;19:2325958219899305.
11. Akatukwasa C, Getahun M, El Ayadi AM, Namanya J, Maeri I, Itiakorit H, Owino L, Sanyu N, Kabami J, Ssemmondo E, Sang N, Kwarisiima D, Petersen ML, Charlebois ED, Chamie G, Clark TD, Cohen CR, Kanya MR, Bukusi EA, Havlir DV, Camlin CS. Dimensions of HIV-related stigma in rural communities in Kenya and Uganda at the start of a large HIV 'test and treat' trial. *PLoS One*. 2021;16(5):e0249462.
12. Nji KE, Nsagha DS, Siysi VV, Ge E orock, Marcelin NN. Knowledge and Attitude Regarding HIV/AIDS and Universal Testing and Treatment Strategy in some Selected Communities in Fako Health Districts of Cameroon. *Journal of Environmental Science and Public Health [Internet]*. 2020 Aug 13 [cited 2025 May 9];4(3):197–215. Available from: <http://www.fotunejournals.com/knowledge-and-attitude-regarding-hiv-aids-and-universal-testing-and-treatment-strategy-in-some-selected-communities-in-fako-health.html>
13. Richard AK, Roland YK, Christian YK, Cécile KKA, Michel AJ, Lacina C, Vincent AK. Knowledge, Attitudes, and Practices of HIV-Positive Adolescents Related to HIV/AIDS Prevention in Abidjan (Côte d'Ivoire). *International Journal of Pediatrics [Internet]*. 2020 [cited 2025 May 9];2020(1):8176501. Available from: <https://onlinelibrary.wiley.com/doi/abs/10.1155/2020/8176501>
14. Abebe M, Asgedom YS, Gebrekidan AY, Wondimagegne YA, Hareru HE, Tebeje TM. Factors associated with HIV testing among young women in Tanzania: Insights from the 2022 Tanzanian Demographic and Health Survey using Anderson's Behavioral Model. *Front Public Health*. 2024;12:1518314.
15. Adamu DJ, Dodor A, Badiekgang SM, Ziem BA, Mohammed AG, Borlabi SB, Bagyiiko BK, Dapaa S, Agyabeng K, Issahaku GR, Billy J, Boakye-Boateng S. Awareness of HIV Status and Barriers to Accessing Testing Services among the Youth in Asutifi North District, Ghana, 2020. *Journal of Interventional Epidemiology and Public Health [Internet]*. 2024 Dec 18 [cited 2025 May 4];7(56). Available from: <https://www.afenet-journal.net/content/article/7/56/full/>
16. Fana T. Knowledge, Attitude and Practices Regarding HIV and AIDS among High School Learners in South Africa. *The Open AIDS Journal [Internet]*. 2021 [cited 2025 May 9]; Available from: <https://openajournal.com/VOLUME/15/PAGE/84/FULLTEXT/>
17. Musheke M, Ntalasha H, Gari S, Mckenzie O, Bond V, Martin-Hilber A, Merten S. A systematic review of qualitative findings on factors enabling and deterring uptake of HIV testing in Sub-Saharan Africa. *BMC Public Health [Internet]*. 2013 Mar 11 [cited 2025 May 5];13(1):220. Available from: <https://doi.org/10.1186/1471-2458-13-220>
18. CAMPHIA. Cameroon [Internet]. PHIA Project. 2024 [cited 2025 May 19]. Available from: <https://phia.icap.columbia.edu/countries/cameroon/>
19. Nkum CB, Nanfak A, Guenou E, Kami RF, Bashombwa AM, Tchio-Nighie KH, Nangue C, Ateudjieu J. Factors Associated with HIV Testing Uptake in Cameroon: Data from the 2018 Cameroon Demographic and Health Survey. *HIV [Internet]*. 2025 Jan 22 [cited 2025 Jan 22];17:9–17. Available from: <https://www.dovepress.com/factors-associated-with-hiv-testing-uptake-in-cameroon-data-from-the-2-peer-reviewed-fulltext-article-HIV>
20. Atanga PN, Ndetan HT, Achidi EA, Meriki HD, Hoelscher M, Kroidl A. Retention in care and reasons for discontinuation of lifelong antiretroviral therapy in a cohort of Cameroonian pregnant and breastfeeding HIV-positive women initiating 'Option B+' in the South West Region. *Tropical Medicine & International Health [Internet]*. 2017 [cited 2025 May 19];22(2):161–70. Available from: <https://onlinelibrary.wiley.com/doi/abs/10.1111/tmi.12816>
21. Kumba. In: Wikipedia [Internet]. 2024 [cited 2025 Jan 17]. Available from: <https://en.wikipedia.org/w/index.php?title=Kumba&oldid=1259323662>
22. Bekolo CE, Kouanfack C, Ateudjieu J, Bechem E, Ndeso S, Tendengfor N, Nsagha D, Choukem S. The declining trend in HIV prevalence from population-based surveys in Cameroon between 2004 and 2018: myth or reality in the universal test and treat era? *BMC Public Health [Internet]*. 2023 Mar 13 [cited 2025 Jan 22];23(1):479. Available from: <https://doi.org/10.1186/s12889-023-15374-8>
23. Wan F, Colditz GA, Sutcliffe S. Matched Versus Unmatched Analysis of Matched Case-Control Studies. *Am J Epidemiol*. 2021 Sep 1;190(9):1859–66.
24. Mansy W, Syed W, Babelghaith S, Al-Arifi MN. Assessment of the knowledge, attitudes, and risk perceptions of pharmacy students regarding HIV/AIDS. *Front Public Health [Internet]*. 2025 Jan 8 [cited 2025 May 9];12:1499898. Available from: <https://www.frontiersin.orghttps://www.frontiersin.org/journal/public-health/articles/10.3389/fpubh.2024.1499898/full>
25. Shamu S, Khupakonke S, Farirai T, Slabbert J, Chidarikire T, Guloba G, Nkhwashu N. Knowledge, attitudes and practices of young adults towards HIV prevention: an analysis of baseline data from a community-based HIV prevention intervention study in two high HIV burden districts, South Africa. *BMC Public Health [Internet]*. 2020 Aug 17 [cited 2025 Jan 16];20(1):1249. Available from: <https://doi.org/10.1186/s12889-020-09356-3>
26. Bujang MA, Omar ED, Baharum NA. A Review on Sample Size Determination for Cronbach's Alpha Test: A Simple Guide for Researchers. *Malays J Med Sci [Internet]*. 2018 Nov [cited 2025 Jun 28];25(6):85–99. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6422571/>
27. Al-Osail AM, Al-Sheikh MH, Al-Osail EM, Al-Ghamdi MA, Al-Hawas AM, Al-Bahussain AS, Al-Dajani AA. Is Cronbach's alpha sufficient for assessing the reliability of the OSCE for an internal medicine course? *BMC Research Notes [Internet]*. 2015 Oct 19 [cited 2025 Jun 28];8(1):582. Available from: <https://doi.org/10.1186/s13104-015-1533-x>
28. Tavakol M, Dennick R. Making sense of Cronbach's alpha. *Int J Med Educ*. 2011 Jun 27;53–5.
29. World Medical Association. Declaration of Helsinki – WMA – The World Medical Association [Internet]. 2025 [cited 2025 May 10]. Available from: <https://www.wma.net/what-we-do/medical-ethics/declaration-of-helsinki/>
30. World Medical Association. World Medical Association Declaration of Helsinki: ethical principles for medical research involving human subjects. *JAMA*. 2013 Nov 27;310(20):2191–4.
31. Elm E von, Altman DG, Egger M, Pocock SJ, Gøtzsche PC, Vandenbroucke JP, Initiative for the S. The Strengthening of Reporting of Observational Studies in Epidemiology (STROBE) Statement: Guidelines for Reporting Observational Studies. *PLOS Medicine [Internet]*. 2007 Oct 16 [cited 2025 May 10];4(10):e296. Available from: <https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.0040296>
32. Ryan S, Hahn E, Rao A, Mwinnyaa G, Black J, Maharaj R, Mvandaba N, Nyanisa Y, Quinn TC, Hansoti B. The impact of HIV knowledge and attitudes on HIV testing acceptance among patients in an emergency department in the Eastern Cape, South Africa. *BMC Public Health [Internet]*. 2020 Jul 6 [cited 2025 Jan 16];20(1):1066. Available from: <https://doi.org/10.1186/s12889-020-09170-x>
33. Karimi K, Azizpour Y, Shafaati M, Nejad SM, Ehsani R, Nikfarjam A, Mousavi ME, Akbarpour S. Assessing HIV transmission knowledge and rapid test history among the general population in Iran. *Sci Rep [Internet]*. 2025 Feb 10 [cited 2025 Apr 29];15(1):4944. Available from: <https://www.nature.com/articles/s41598-025-88844-1>
34. Takahashi TA, Johnson KM, Bradley KA. A Population-Based Study of HIV Testing Practices and Perceptions in 4 U.S. States. *J Gen Intern Med [Internet]*. 2005 Jul [cited 2025 May 10];20(7):618–22. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1490154/>
35. Maduakolam IO, Ogbonnaya NP, Ndubuisi IF, Mbadugha CJ. Knowledge of HIV/AIDS, Attitude and use of Voluntary Counselling and Testing among Adolescents in Senior Secondary Schools, Enugu North Local Government Area, Enugu State. *Journal of Nursing and Women's Health [Internet]*. 2020 Mar 2 [cited 2025 May 19]; Available from: <https://www.gavinpublishers.com/article/view/knowledge-of-hiv-aids-attitude-and-use-of-voluntary-counselling-and-testing-among-adolescents-in-senior-secondary-schools-enugu-north-local-government-area-enugu-state>
36. Ndlovu SMS, Ross A, Ndirangu J. Knowledge of HIV and/or AIDS and HIV testing services among young men in South Africa. *African Journal of Primary Health Care & Family Medicine [Internet]*. 2023 Jul 31 [cited 2025 Apr 28];15(1):10. Available from: <https://phcfm.org/index.php/phcfm/article/view/3796>
37. Dzah SM, Tarkang EE, Lutala PM. Knowledge, attitudes and practices regarding HIV/AIDS among senior high school students in Sekondi-Takoradi metropolis, Ghana. *Afr J Prim Health Care Fam Med [Internet]*. 2019 May 9 [cited 2025 May 20];11(1):1875. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6556927/>
38. Quarm MD, Mthembu J, Zuma K, Tarkang EE. Knowledge, attitudes and prevention practices regarding HIV/AIDS among barbers in Ho municipality, Ghana. *SAHARA J*. 2021 Dec;18(1):42–51.
39. Tsala Dimbuene Z, Kuete Defo B. Fostering accurate HIV/AIDS knowledge among unmarried youths in Cameroon: Do family environment and peers matter? *BMC Public Health [Internet]*. 2011 May 19 [cited 2025 May 19];11(1):348. Available from: <https://doi.org/10.1186/1471-2458-11-348>
40. Siddique AB, Maruf MdFI, Sakib MR, Hasan M, Hossain M, Paul DG, Tasnim MostN, Akter S, Debnath B, Rahman T, Rahman M. Assessment of knowledge, attitudes, and prevention practices regarding HIV/AIDS among urban slum dwellers in Bangladesh: an interview-based study. *BMC Public Health [Internet]*. 2025 Feb 25 [cited 2025 May 9];25(1):775. Available from: <https://doi.org/10.1186/s12889-025-21990-3>

41. Katirayi L, Tchendjou P, Tchounga B, Mbunka M, Wicks M, Conserve DF. Changing attitudes towards HIV testing and treatment among three generations of men in Cameroon: a qualitative analysis using the Fogg Behavior Model. *BMC Public Health* [Internet]. 2023 Mar 15 [cited 2025 May 19];23(1):501. Available from: <https://doi.org/10.1186/s12889-023-15139-3>
42. Dzinamarira T, Moyo E. Adolescents and young people in sub-Saharan Africa: overcoming challenges and seizing opportunities to achieve HIV epidemic control. *Front Public Health* [Internet]. 2024 Mar 19 [cited 2025 May 19];12. Available from: <https://www.frontiersin.org/journals/public-health/articles/10.3389/fpubh.2024.1321068/full>
43. Moyo E, Moyo P, Murewanhema G, Mhango M, Chitungo I, Dzinamarira T. Key populations and Sub-Saharan Africa's HIV response. *Front Public Health* [Internet]. 2023 May 16 [cited 2025 Jan 22];11:1079990. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10229049/>
44. Sabo KG, Seifu BL, Kase BF, Asebe HA, Asgedom YS, Shibeshi AH, Tebeje TM, Lombebo AA, Fente BM, Mare KU. Factors influencing HIV testing uptake in Sub-Saharan Africa: a comprehensive multi-level analysis using demographic and health survey data (2015–2022). *BMC Infectious Diseases* [Internet]. 2024 Aug 13 [cited 2025 May 19];24(1):821. Available from: <https://doi.org/10.1186/s12879-024-09695-1>
45. Ndenkeh JJN, Nelson LE, Bogning A, Yumo H, Bediang G, Njih E, Fotso D, Abu-Ba'are GR, Kouanfack C, Ni Z. The Feasibility and Acceptability of Using Mobile Applications for HIV Testing, Prevention, and Treatment Among Adolescent Girls and Young Women in Cameroon: A Cross-Sectional Study. *J Assoc Nurses AIDS Care*. 2024 Jun 1;35(3):210–21.
46. Ajayi AI, Awopegba OE, Adeagbo OA, Ushie BA. Low coverage of HIV testing among adolescents and young adults in Nigeria: Implication for achieving the UNAIDS first 95. *PLoS One* [Internet]. 2020 May 19 [cited 2025 May 19];15(5):e0233368. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7237011/>
47. Schwarcz S, Richards TA, Frank H, Wenzel C, Hsu LC, Chin CSJ, Murphy J, Dilley J. Identifying barriers to HIV testing: personal and contextual factors associated with late HIV testing. *AIDS Care*. 2011 Jul;23(7):892–900.
48. WHO. WHO publishes new Consolidated HIV guidelines for prevention, treatment, service delivery & monitoring [Internet]. 2021 [cited 2025 Mar 14]. Available from: <https://www.who.int/news/item/16-07-2021-who-publishes-new-consolidated-hiv-guidelines-for-prevention-treatment-service-delivery-monitoring>

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.